A Mersenne-Prime Dichotomy in Pascal's First Divisible Row

A precise two-prime family for OEIS A349958

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Abstract

Let $a(N)$ denote the first row index of Pascal's triangle containing a binomial coefficient divisible by N (OEIS A349958). We determine $a(2p)$ exactly for every odd prime p: it equals $p+1$ if p is a Mersenne prime, and p otherwise. This confirms the OEIS prime-power fixed-point conjecture on the infinite family $N = 2p$ and identifies a finer exceptional dichotomy. It does not establish the conjecture for arbitrary composites. The argument is self-contained apart from the elementary binary binomial congruence, which is proved below.

The OEIS problem and exact statement

OEIS A349958 defines, for $N \geq 2$,

$$a(N) = \min\{j \geq 0 : \exists\, 0 \leq k \leq j,\ N \mid \tbinom{j}{k}\}.$$

The entry records the conjecture $a(N) = N$ if and only if N is a prime power, together with the established prime-power case. We prove an explicit family of the non-prime-power direction. Throughout, a *Mersenne prime* means a prime of the form $2^t - 1$ with $t \geq 2$.

Theorem 1. *For every odd prime p,*

$$\boxed{a(2p) = \begin{cases} p+1, & p = 2^t - 1 \text{ for some } t \geq 2, \\ p, & \text{otherwise.} \end{cases}}$$

In particular, $a(2p) < 2p$ for every odd prime p.

Lemma 1 (Binary binomial criterion)**.** *For $m \geq 1$, every coefficient $\binom{m}{k}$, $0 \leq k \leq m$, is odd if and only if $m = 2^t - 1$ for some $t \geq 1$. Otherwise, there is an integer k with $0 < k < m$ for which $\binom{m}{k}$ is even.*

Proof. Write $m = \sum_{i=0}^{t-1} \varepsilon_i 2^i$ with $\varepsilon_{t-1} = 1$ and $\varepsilon_i \in \{0, 1\}$. Over $\mathbb{F}_2[X]$, repeated squaring gives

$$(1+X)^m = \prod_{i:\, \varepsilon_i = 1} (1 + X^{2^i}).$$

Every subset of the present binary digits has a distinct sum, so the coefficient of X^k is 1 precisely when the 1-bits of k are among the 1-bits of m. If all digits below the leading digit are 1, then $m = 2^t - 1$ and every $0 \leq k \leq m$ appears. Otherwise choose a missing digit $i < t - 1$ and $k = 2^i$; then $0 < k < m$ and its coefficient is even. □

Proof of Theorem 1. If $j < p$, the integer $j!$ is not divisible by p, so none of its binomial coefficients is divisible by p. Thus $a(2p) \geq p$. For $1 \leq k \leq p - 1$,

$$\binom{p}{k} = \frac{p}{k}\binom{p-1}{k-1}$$

is divisible by p, since $p \nmid k$. Because p is odd, $2p$ divides $\binom{p}{k}$ exactly when this coefficient is even.

If p is not of the form $2^t - 1$, Lemma 1 supplies such an interior even coefficient; hence $a(2p) = p$.

If $p = 2^t - 1$ is prime (necessarily $t \geq 2$), all coefficients in row p are odd, so $a(2p) > p$. In row $p + 1$,

$$\binom{p+1}{2} = \frac{p(p+1)}{2} = p\,2^{t-1},$$

which is divisible by $2p$. Therefore $a(2p) = p + 1$. Finally, $p + 1 < 2p$ for $p \geq 3$. □

Checks, scope, and precedence

The formula yields $a(6) = 4$, $a(10) = 5$, $a(14) = 8$, $a(22) = 11$, and $a(62) = 32$, matching the published terms of A349958. An independent exact-integer computation checked it for every odd prime $p < 100$; the proof is valid without this check. This result treats only $N = 2p$ and does not resolve the full OEIS conjecture. The binary criterion is classical (a special case of Lucas's theorem). No claim of mathematical novelty or priority is made.

References and disclosure

1. OEIS Foundation, *A349958: First row containing a binomial coefficient divisible by n*, https://oeis.org/A349958 (accessed 2026-11-14).
2. E. Lucas, classical binomial-coefficient congruence modulo a prime (Lucas's theorem); the needed case modulo 2 is proved above.

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